

Lists in Python – Complete Tutorial

Lists in Python – Complete Tutorial Notebook

This notebook is a **complete guide to Lists in Python**, covering theory, examples, loops, string integration, list comprehensions, nested lists, practical use-cases, and logical practice exercises.

1. Introduction to Lists

```
numbers = [10, 20, 30]
names = ['Ali', 'Sara', 'Ahmed']
mixed = [1, 'Python', 3.5, True]
numbers, names, mixed
```

2. Creating Lists

```
empty1 = []
empty2 = list()
marks = [85, 90, 78]
empty1, empty2, marks
```

3. Accessing Elements

```
fruits = ['apple', 'banana', 'cherry']
fruits[0], fruits[-1]
```

4. Slicing Lists

```
nums = [0,1,2,3,4,5]
nums[1:4], nums[:3], nums[::2]
```

5. Modifying Lists

```
fruits.append('orange')
fruits.insert(1, 'mango')
fruits.extend(['kiwi', 'melon'])
fruits
```

6. Removing Elements

```
fruits.remove('banana')
removed = fruits.pop()
fruits, removed
```

7. List Functions and Methods

```
lst = [1,2,2,3]
lst.append(4)
lst.insert(1,10)
lst.extend([5,6])
lst.remove(2)
lst.count(2), lst.index(3)
```

```
lst.sort()
lst.reverse()
copy_lst = lst.copy()
len(lst), sum(lst), min(lst), max(lst), copy_lst
```

8. Lists with Loops

```
for item in lst:
    print(item)
```

9. Lists with String Functions

```
names = ['ali', 'ahmed', 'sara']
[name.upper() for name in names]
```

10. List Comprehension

```
squares = [x*x for x in range(1,6)]  
evens = [x for x in range(1,21) if x%2==0]  
squares, evens
```

11. Nested Lists

```
matrix = [[1,2,3],[4,5,6]]  
for row in matrix:  
    for item in row:  
        print(item, end=' ')  
    print()
```

12. Practical Examples

```
marks = [80,75,90,85]  
sum(marks)/len(marks)
```

```
sentence = 'Python is very powerful'  
words = sentence.split()  
[len(word) for word in words]
```

13. Practice Exercises

1. Find unique elements in a list
2. Longest and shortest word
3. Frequency counter
4. Sentence analyzer
5. Rotate list by n
6. Student result processing
7. Word pair generator